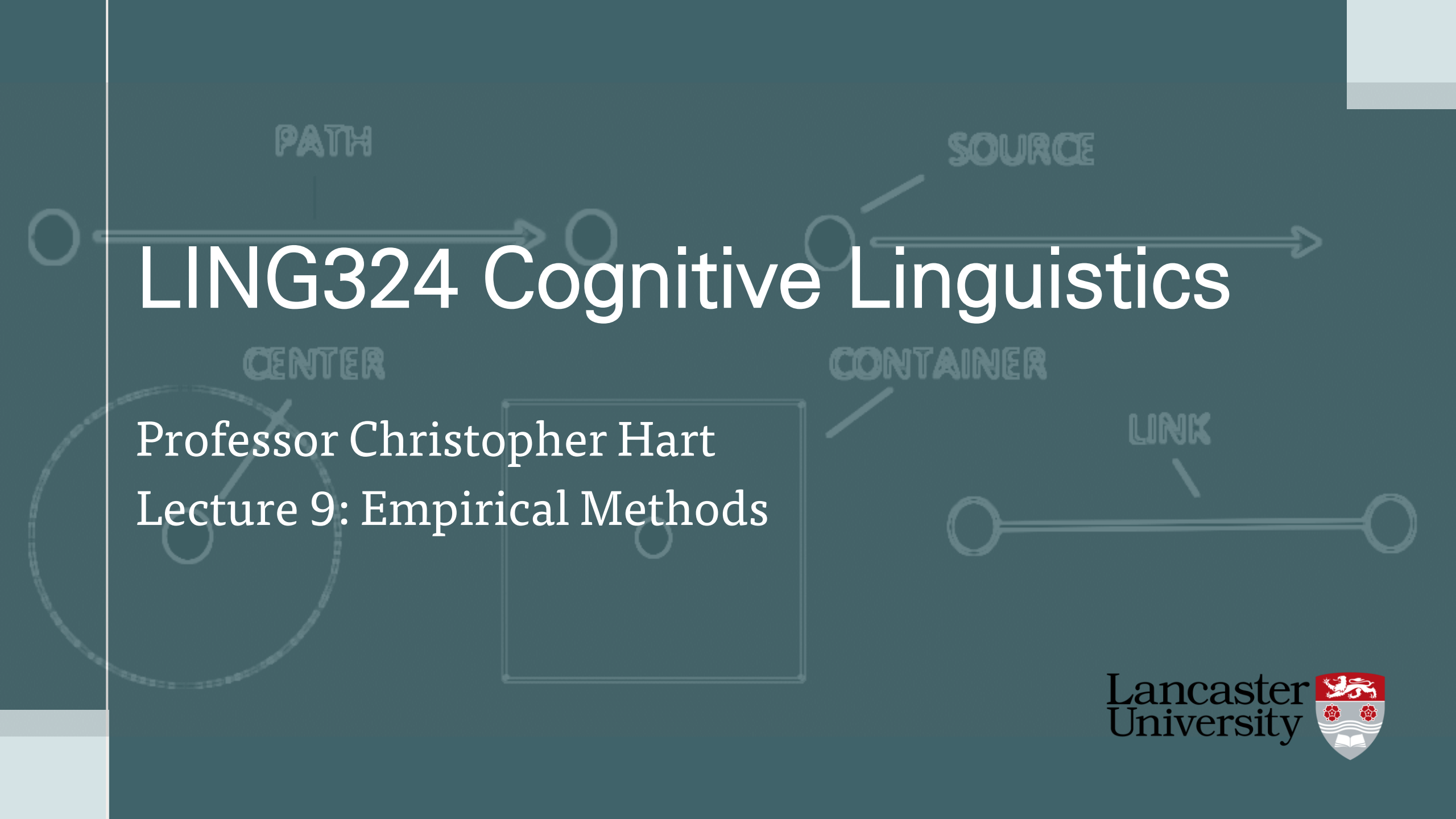




LING324 Cognitive Linguistics

Professor Christopher Hart
Lecture 9: Empirical Methods

Overview

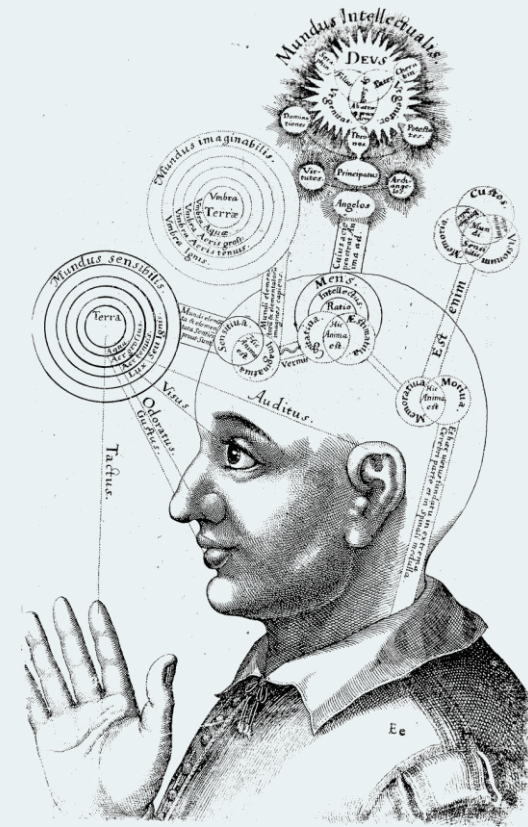
- Introduction
- Simulation
- Encyclopaedic knowledge
- Perspective
- Dynamicity
- Metaphor
- A word of caution ...

Introduction

- **Introspection** main method of cognitive linguistics

The method of introspection can be justified in much the same way as the methods settled on by any science. In any science, a researcher must go to where the relevant data under study are to be found. For example, if one's area of scientific study is geology, one must go examine the earth. Here, "going to where the data are" entails physical travel to terrestrial sites. In the same way, if one's area of scientific study is linguistic meaning, one must go to where meaning is located. And meaning is located in conscious experience. In the case of such subjective data, "going" to their location consists of introspection. (Talmy

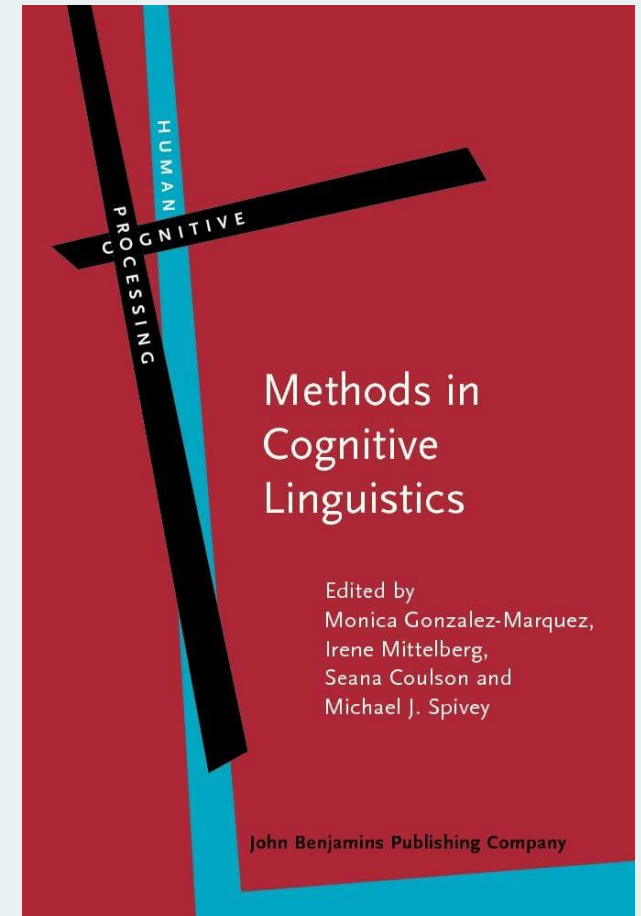
2000a: 5-6)



Introduction

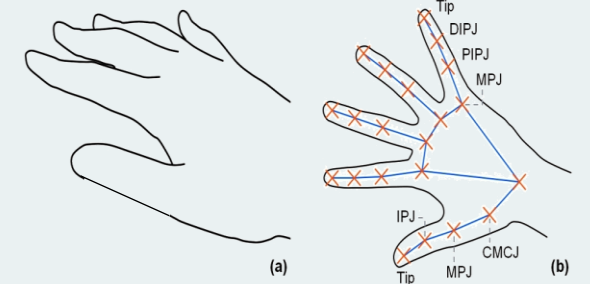
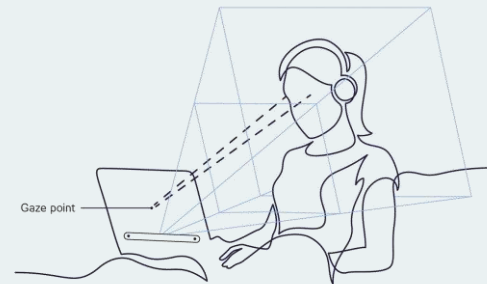
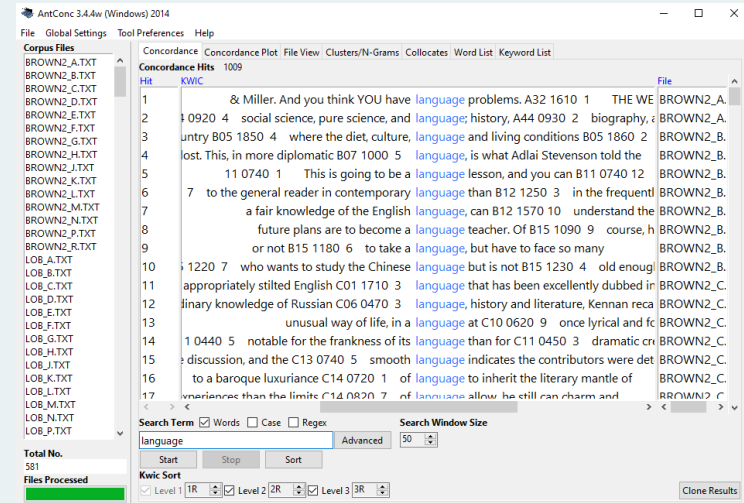
- CL committed to psychological reality of theoretical constructs
- However, growing awareness that “linguistic theory should be grounded in observation of language usage, in experimental tests of its validity, and in general knowledge of cognitive function” (Gonzalez-Marquez et al. 2007: xxii)

Because cognitive linguistics does not assume isolation for linguistic processes, most of the methods used to study cognition can potentially be adapted to investigating language. At the same time, language usage data, including co-speech gesture and signed languages, are sources of overt manifestations of cognitive processing. (Gonzalez-Marquez et al. 2007: xxii)



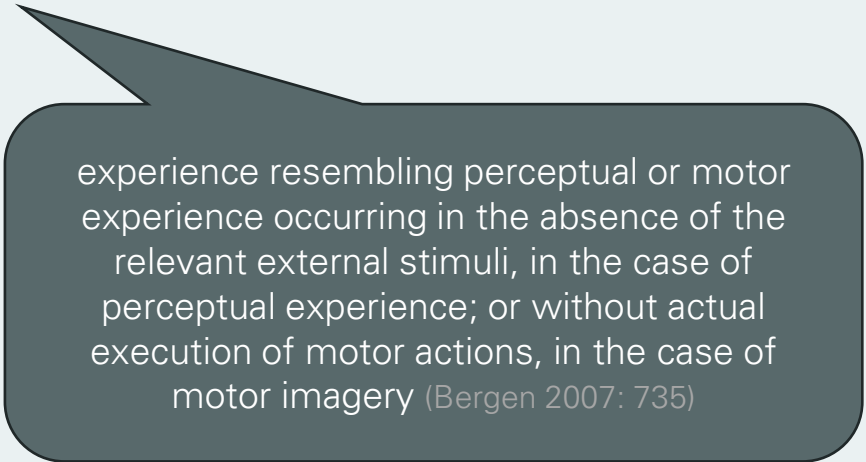
Introduction

- Three lines of evidence:
 - Corpus linguistics
 - Behavioural experiments
 - Cross-modal priming effects
 - Eye-tracking
 - Neuroimaging
 - Gesture analysis



Simulation

- Behavioural and gestural methods reveal **simulations**
 - Mental “re-enactments” of experience
 - Perceptual vs. motor
 - Meaning not propositional but modal (mental imagery)
 - Size, shape, orientation, colour, weight, smell, texture ...
 - Speed, force, direction ...
 - Necessarily also viewpointed
- Supports and enhances domain-general, embodied view of language



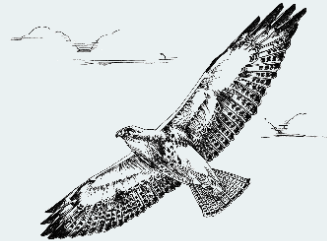
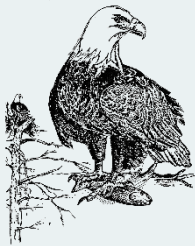
experience resembling perceptual or motor experience occurring in the absence of the relevant external stimuli, in the case of perceptual experience; or without actual execution of motor actions, in the case of motor imagery (Bergen 2007: 735)

Encyclopaedic Knowledge

- Meaning includes **world knowledge**
 - Meaning, in the form of mental imagery, sensitive to context
- Cross-modal priming effects
 - **Image-sentence compatibility effects (ICE)**
 - Subjects respond faster in trials where picture and mental image correspond compared to trials where do not
 - Sentence-first vs. picture-first judgement task

Encyclopaedic Knowledge

- (a) The ranger saw the eagle in its nest
- (b) The ranger saw the eagle in the sky



- Exp.1: Was the object mentioned in the preceding sentence?
- Exp. 2: Name the object

Table 1. Object recognition latencies and accuracy in Experiment 1 and picture naming times in Experiment 2

Measure	Condition		
	Match	Mismatch	Neutral
Experiment 1			
Reaction time	697 (202)	761 (210)	—
Percentage correct	97 (6)	93 (7)	—
Experiment 2			
Reaction time	605 (115)	638 (128)	617 (125)

Note. Standard deviations are given in parentheses.

Subjects represented the implied shape of the object when comprehending the sentence, so that responses to the picture were slower when the picture mismatched the implied shape than when there was a match between the pictured and implied shapes (p.169)

Encyclopaedic Knowledge

- (a) Mary hammered the nail into the wall
- (b) Mary hammered the nail into the floor



- Was the object mentioned in the preceding sentence?

Participants responded more quickly ($M = 838$, $SD = 331$ vs. $M = 882$, $SD = 329$) when the picture matched the implied orientation of the sentence than when it did not, $F(1, 36) = 6.36$, $p < .05$.

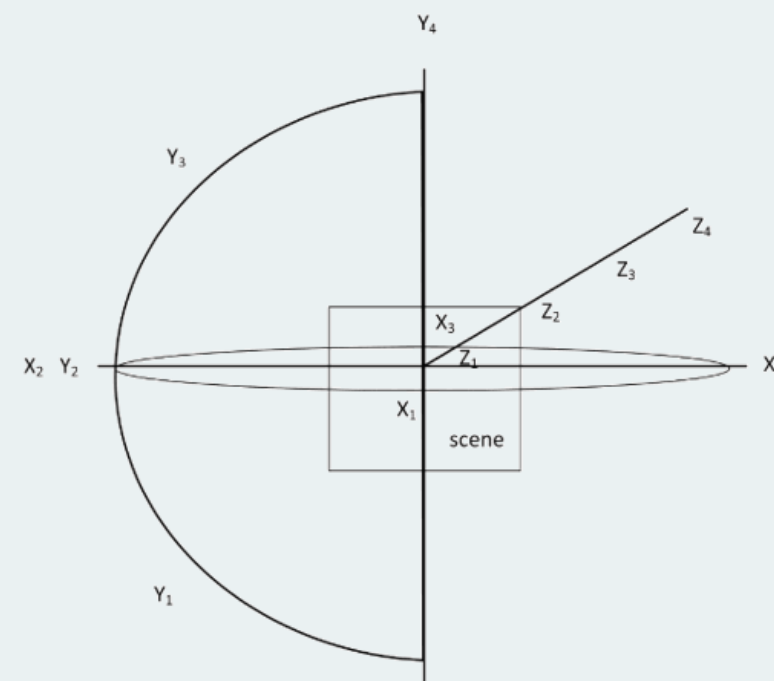
The recognition of an object mentioned in a sentence was influenced by the orientation of the object. Specifically, recognition latencies were shorter for pictures of objects whose orientation matched the orientation suggested by the prior text than for pictures of objects whose orientation did not match the one suggested by the prior text. (p.155)

Perspective

- Meaning includes **perspective**
 - Viewpoint from which conceptual content construed

Many expressions undeniably invoke a vantage point as part of their meaning (arguably, all expressions do) (Langacker 2008: 75)

- Dimensions of perspective: internal vs. external, anchor, angle, distance
- Implicated in meaning of locative PPs, deictic expressions, verbs and perfective vs. imperfective aspect

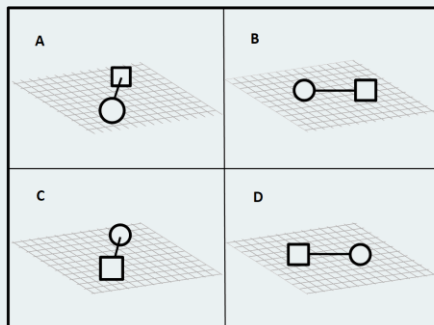


(Hart 2015)

Perspective

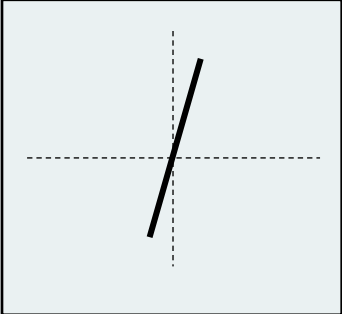
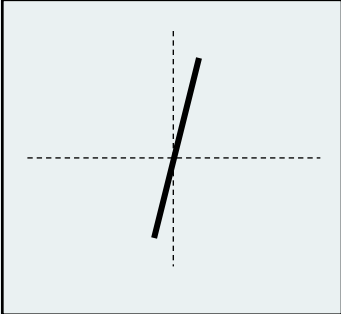
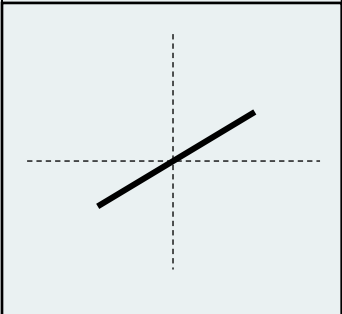
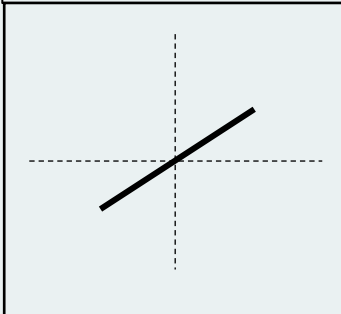
- Transitive vs. reciprocal verbs
- Sentence-image matching task

- a) The circle attacked the square
- b) The circle clashed with the square



Sagittal = 0°
Transversal = 90°

Mean axis angles

	
○ VERBED □	□ was VERBED by ○
	
○ VERBED with □	□ VERBED with ○

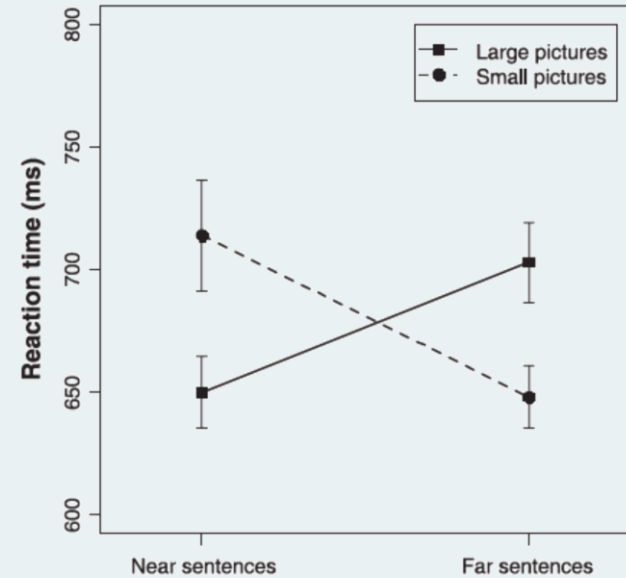
(Hart 2019)

Perspective

- (a) You are looking at a fire hydrant right in front of you
- (b) You are looking at a fire hydrant from afar



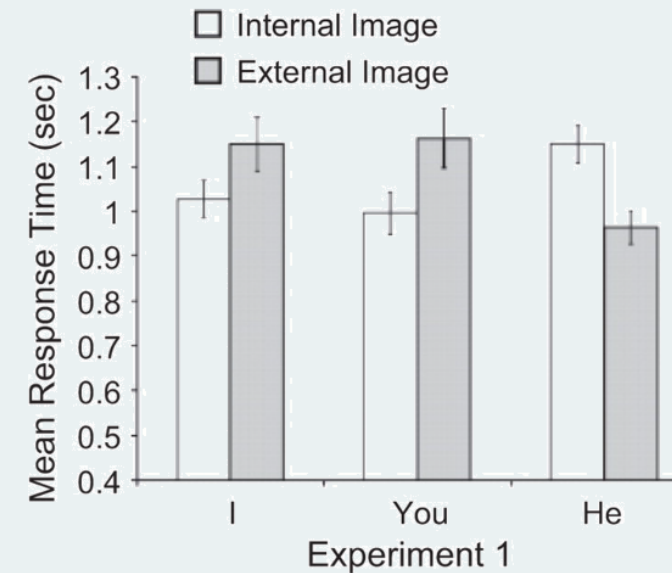
- Was the object mentioned in the preceding sentence?



When reading sentences about distant objects, comprehenders simulate smaller objects, and when they read sentences about nearby objects, they simulate larger objects (p.7)

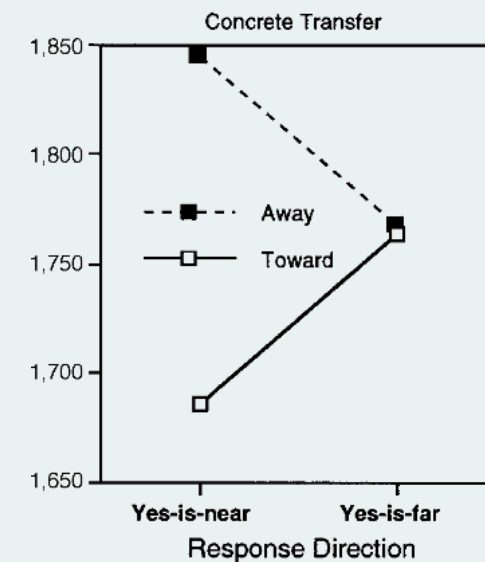
Perspective

- Actor vs. onlooker perspective
- Modulated by pronouns
 - a) I am slicing the tomato
 - b) You are slicing the tomato
 - c) He is slicing the tomato
- Does the picture match the event described?



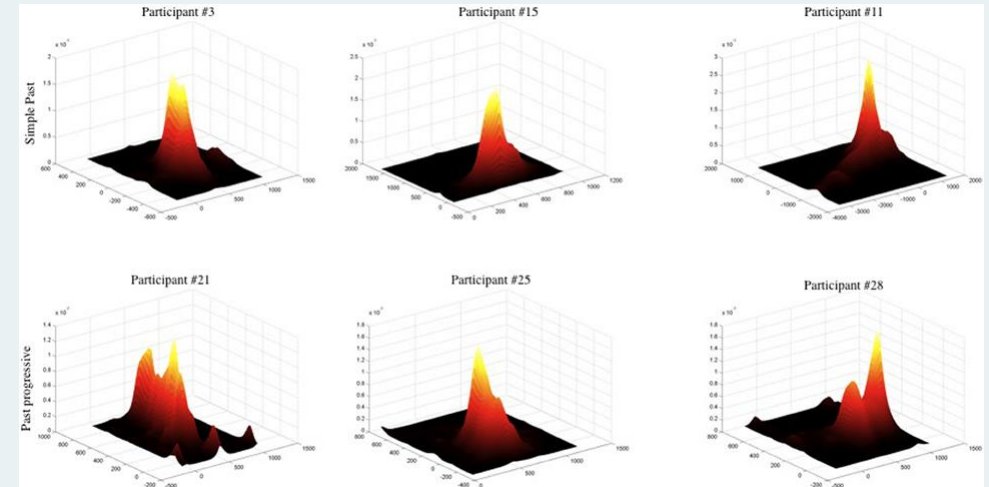
Perspective

- Action-sentence compatibility effects (ACE)
 - Motor simulations
 - Subjects faster in trials where action and imagined action consistent compared to trials where not
 - Sentence-first vs. action first
 - a) Courtney handed you the notebook
 - b) You handed Courtney the notebook
 - Does the sentence make sense?



Perspective

- Eye movements follow (or facilitate) mental simulations
- Measured in **eye-tracking** studies
 - Fixations
 - Saccades
- Perfective vs. progressive aspect
 - John was on a bike ride yesterday. After he **sped / was speeding** across the valley, he **climbed / was climbing** a mountain range. Then he **pedalled / was pedalling** along a river and finally, he **coasted / was coasting** into a campground.



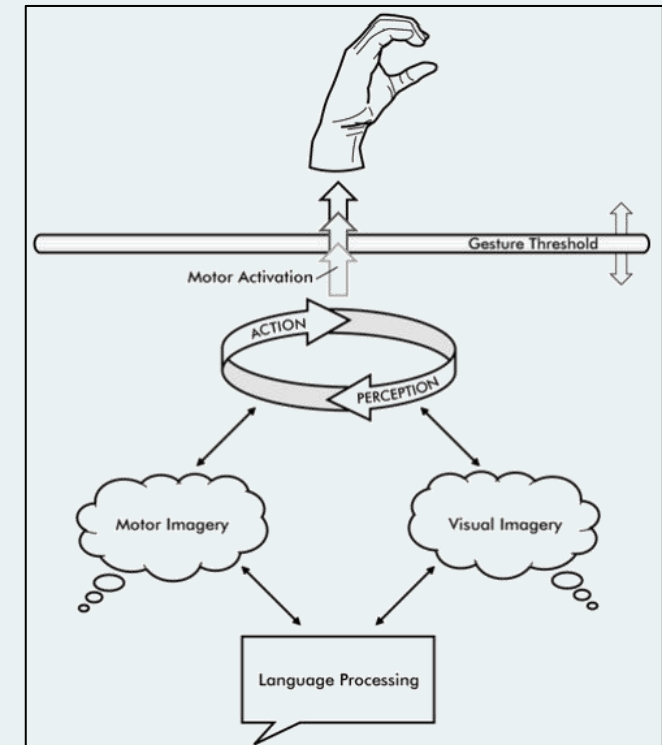
The actual eye movement patterns are in line with what was predicted based on linguistic analyses of aspect and previous experimental work: past progressive appears to emphasize the ongoing motion of described actions and the details of described events, such that sentences with past progressive induce eye movements to be shorter and more dispersed. (p.5)

Perspective

- Gesture as simulated action (Hostetter & Alibali 2008)

gestures emerge from the perceptual and motor simulations that underlie embodied language and mental imagery (p.502)

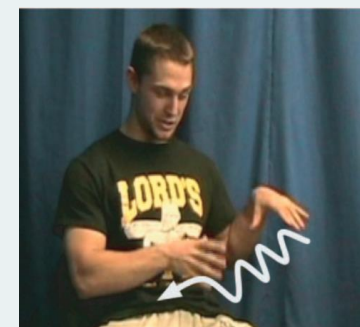
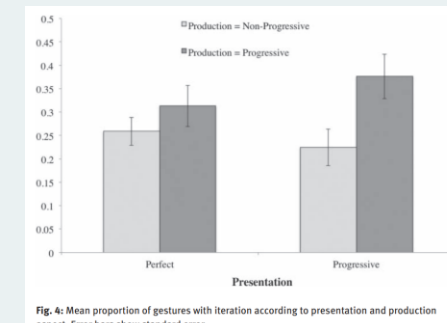
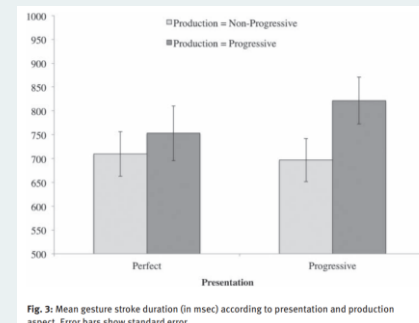
- Gestures reflect spatial properties of mental simulations, including perspective



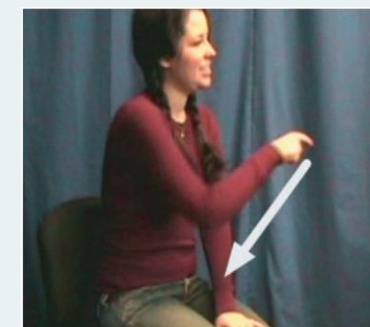
(Hostetter & Alibali 2008)

Perspective

- Perfective vs. progressive aspect
 - There was a trail through the woods. A woman **was hiking/had hiked** down a little hill on the winding path. Afterwards, she **had walked/was walking** up to a little stream. A big stick **was floating/had floated** down the stream in front of a pair of ducks. The ducks **had paddled/were paddling** around it. Then they **were bobbing/had bobbed** around a bend in the stream out of sight. The woman **had leapt/was leaping** along some rocks to the other side of the stream
- When presented in progressive aspect, speakers retelling story used progressive aspect plus longer and more complex gestures reflecting event-internal perspective



Speech: *the stick was floating downstream*

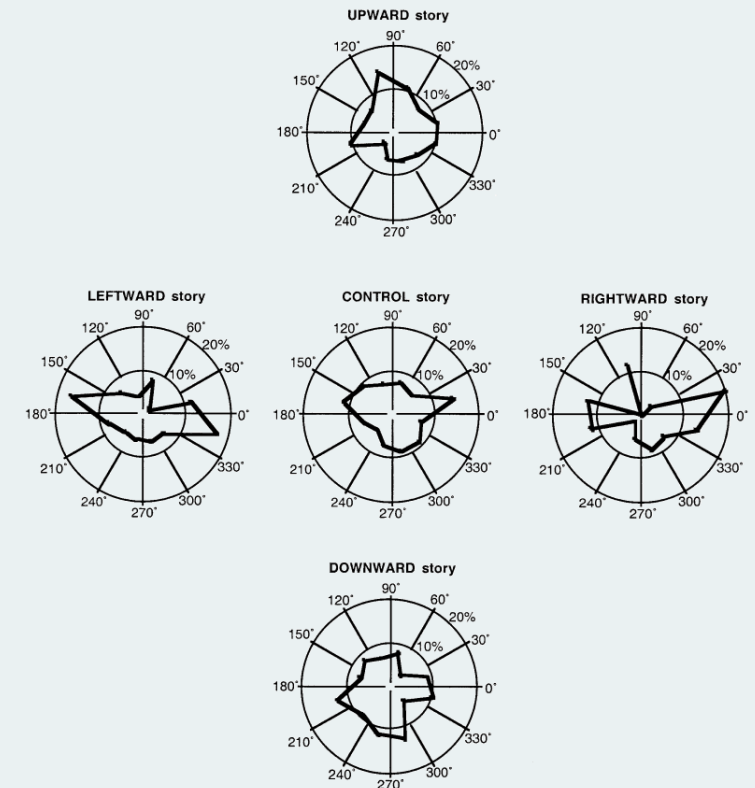


Speech: *it had floated down*

(Parrill, Bergen & Lichtenstein 2013)

Dynamicity

- Conceptualisation involves dynamic construal
 - Narrative
 - Fictive motion
- Reflected in eye-movements
- Upward, downward, leftward, rightward stories
 - Imagine that you are standing across the street from a 40 story apartment building. At the bottom there is a doorman in blue. On the 10th floor, a woman is hanging her laundry out the window. On the 29th floor, two kids are sitting on the fire escape smoking cigarettes. On the very top floor, two people are screaming.

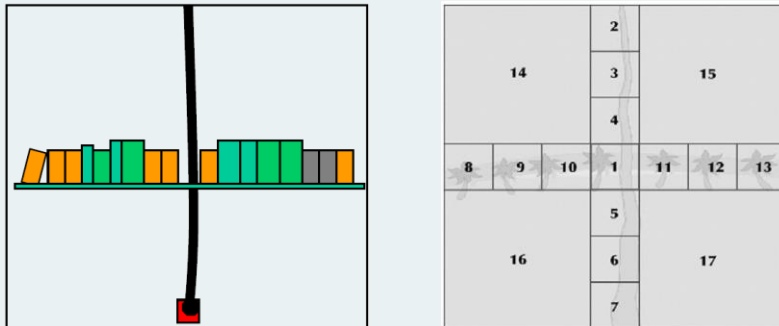


(Spivey & Geng 2001)

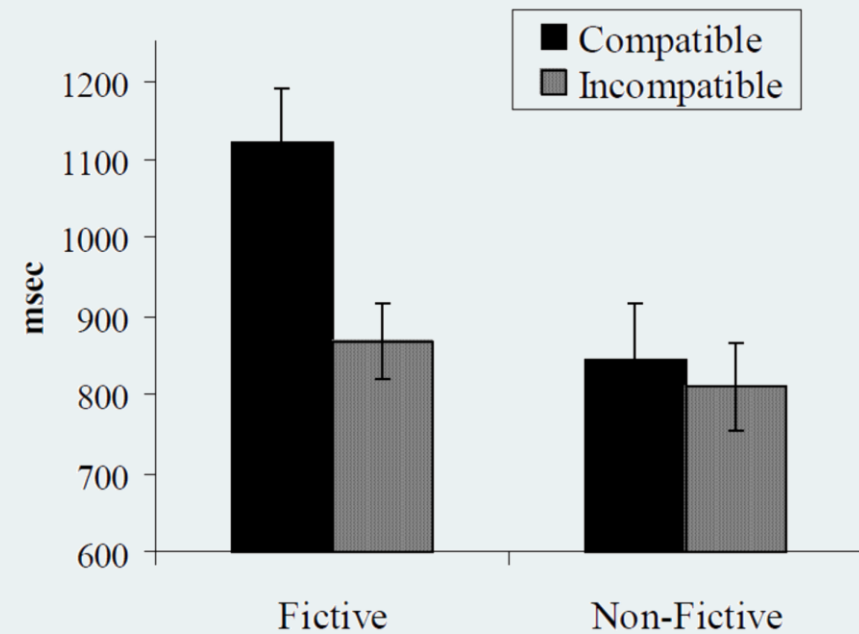
Dynamicity

- a) The books are on the wall
- b) The books run along the wall

- a) The cord is on the wall
- b) The cord runs along the wall

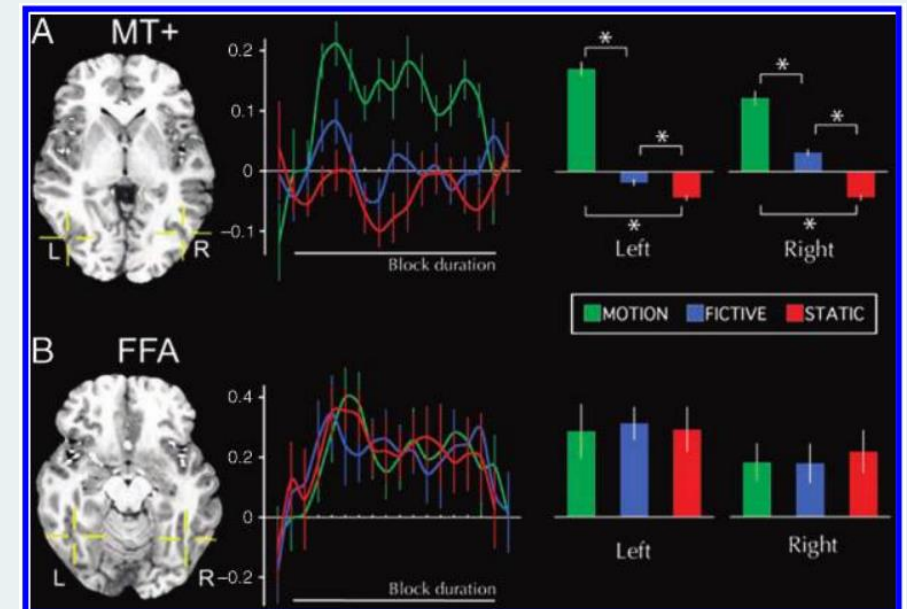


Gaze durations in regions 2-7 and 8-13



Dynamicity

- Functional magnetic resonance imaging (fMRI) shows brain activity during language processing
- Brain areas associated with perception and motor execution activated in sentence comprehension
 - a) I drove from Modesto to Fresno (motion)
 - b) The highway runs from Modesto to Fresno (fictive)
 - c) Modesto and Fresno are in California (static)
- Motion-sensitive visual areas (MT+) activated in motion and fictive motion sentence processing



(Saygin et al. 2009)

Metaphor

- Metaphor interpretation as embodied simulation (Gibbs 2006)

Our ability to make sense of metaphorical language ... resides in the automatic construction of a simulation whereby we imagine performing the bodily actions referred to in the language. Thus, understanding metaphorical expressions like 'grasp a concept' or 'get over' an emotion involve simulating what it must be like to engage in these specific activities, even though these actions are, strictly speaking, impossible to physically perform. (p.434)

- Action-sentence compatibility effect:
 - Comprehension speed increased when metaphor preceded by corresponding physical action compared to mismatching action and no action

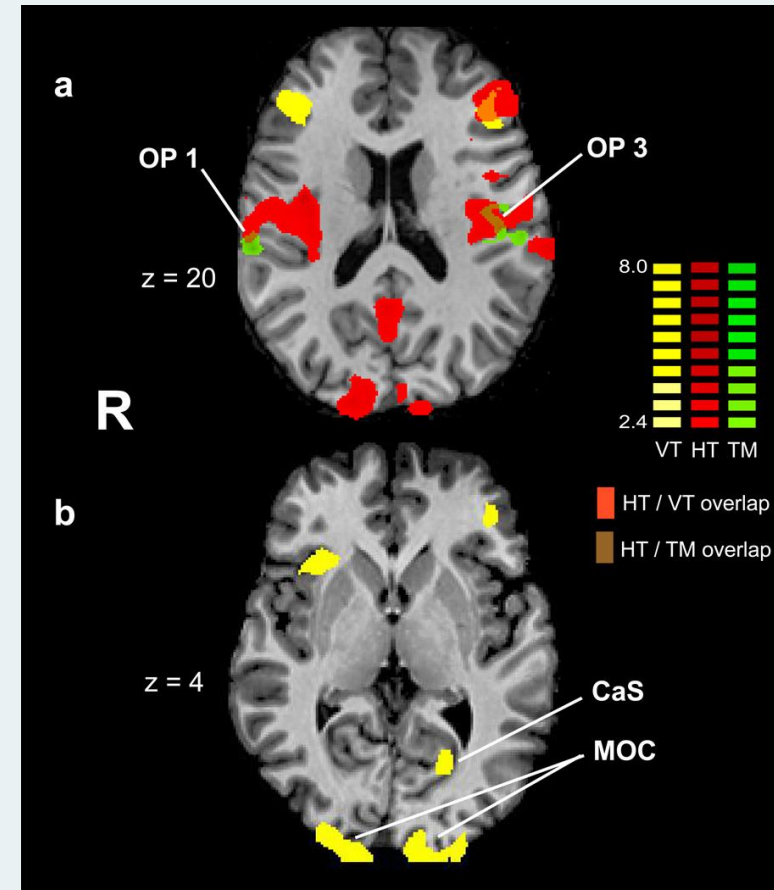
Action	Icon	Metaphorical Phrase
Stamp	^	Stamp out a fear
Push	&	Push the argument
Swallow	#	Swallow your pride
Sniff	*	Sniff out the truth
Spit	:	Spit out the facts
Shake	%	Shake off a feeling
Grasp	“	Grasp a concept
Chew	/	Chew on an idea
Stretch	{	Stretch for understanding

Comprehension times in the matching prime condition ($m = 2629$ milliseconds, $sd = 874$) were significantly faster than in the mismatching prime condition ($m = 3172$ milliseconds, $sd = 864$), $t(48) = 3.3$, $p < 0.01$ and the no prime condition ($m = 2927$ milliseconds, $sd = 687$), $t(48) = 4.4$, $p < 0.001$. The times for the no prime and mismatching prime conditions did not significantly differ. These data support the idea that engaging in appropriate body movements primes people's speeded comprehension of certain metaphorical phrases (i.e., an action metaphor compatibility effect).

Metaphor

- Brain areas associated with source-domain activated when processing metaphor
- E.g. texture-selective somatosensory cortex activated in processing textural metaphors
 - a) She had a bad day
 - b) She had a rough day
- Psychological reality of metaphors with neurological basis

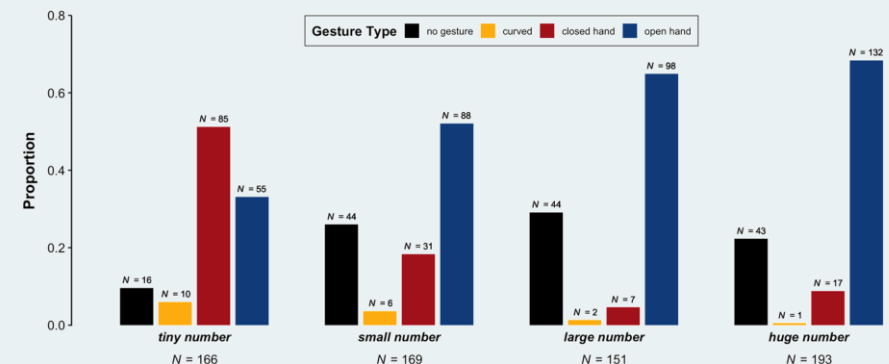
Textural metaphor activations (green) overlaid on same individual's haptic (red) and visual (yellow) texture-selective regions



(Lacey, Stilla & Sathian 2012)

Metaphor

- Source-domain in conceptual metaphors represented iconically in co-speech gestures
- Active figurativity of metaphorical expressions – metaphors very much alive and reflected in gestures (Müller 2008)
- E.g. QUANTITY IS SIZE
 - When referring to greater vs. lesser quantities speakers more likely to gesture (i) with open vs closed hand shape, (2) with outward vs. inward motion, (3) with greater distance between hands



(Winter et al. 2020)

A Word of Caution ...

- Action-sentence compatibility effect (ACE) hallmark finding in favour of embodied language processing
- Glenberg & Kaschak (2002) cited 3218 times as of November 2024 (*Google Scholar*)
- Effect reported in several follow-up studies (e.g. Borreggine & Kaschak 2006; de Vega & Urritia 2011; de Vega et al. 2013; Glenberg et al. 2008; Kaschak & Borreggine 2008; Taylor & Zwaan 2008; Zwaan & Taylor 2006)
- However, recent failures to replicate effect and re-analysis of data in original studies shows only weak support for ACE or support for null hypothesis (Morey et al. 2022; Papesh 2015)

Figure	Article	Experiments	Analysis/Effect	Scaled JZS Bayes Factor	Odds Favor	2:1 Threshold	3:1 Threshold
A	Current	1	ACE interaction	1.96	Alternative	N	N
B	Current*	2	ACE interaction	7.72	Null	N	N
C	Current*	3	ACE interaction	7.64	Null	N	N
D	Current*	4A	ACE interaction	9.03	Null	N	N
E	Current*	4B	ACE interaction	4.06	Null	N	N
F	Current*	5	ACE interaction	9.97	Null	N	N
G	Current*	6	ACE interaction	12.02	Null	N	N
H	Current*	7	ACE interaction	1.71	Null	N	N
I	Current*	8	ACE interaction	10.96	Null	N	N
J	Glenberg & Kaschak (2002)	1	ACE interaction	3.97	Alternative	Y	Y
K	Glenbera & Kaschak (2002)	2A	ACE interaction	1.78	Null	N	N
L	Borreggine & Kaschak (2006)	1	ACE interaction	4.05	Alternative	Y	Y
M	Boulenger et al.(2006)*	2	RT for verbs vs nouns	1.44	Null	N	N
N	Boulenger et al. (2006)	2	Peak wrist acceleration for verbs vs nouns	1.89	Alternative	N	N
O	Zwaan & Taylor (2006)	2	Sentence rotation × Manual rotation	1.27	Null	N	N
P	Zwaan & Taylor (2006)	3	Observed rotaion × Sentence direction	5.97	Alternative	Y	Y
Q	Zwaan & Taylor (2006)	4	Verb region, congruence with rotation direction	2.55	Null	N	N
R	Zwaan & Taylor (2006)	5	Illusory rotation, verb region, congruence with sentence direction	2.70	Null	N	N
S	Glenberg et al. (2008)	1	ACE interaction	1.46	Alternative	N	N
T	Kaschak & Borreggine (2008)	1	500 ms condition	8.72	Alternative	Y	Y
U	Kaschak & Borreggine (2008)	1	toward-toward, 500 ms condition	1.61	Alternative	N	N
V	Kaschak & Borreggine (2008)*	1	away-away, 500 ms condition	2.13	Null	N	N
W	Taylor & Zwaan (2008)	1	Verb region, ACE interaction	2.64	Null	N	N
X	Taylor & Zwaan (2008)	1	Adverb region, ACE interaction	1.32	Null	N	N
Y	Taylor & Zwaan (2008)	2	Verb region, ACE interaction	2.32	Alternative	Y	N
Z	de Vega & Urrutia (2011)	1	Factual sentences, 200 ms delay	1.49	Null	N	N
AA	de Vega et al. (2013)*	1	350 ms verb-cue SOA, motor response time	2.59	Null	N	N
AB	de Vega et al. (2013)	1	350 ms verb-cue SOA, semantic decision time	1.55	Null	N	N
AC	de Vega et al. (2013)	1	350 ms verb-cue SOA, total response time (motor & semantic)	1.13	Null	N	N

(Papesh 2015)

Thank you

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